

MATEO ROBLES

Creative Technologist | Generative Art, WebGL, Interactive AI
Walnut Creek, CA | +1 925 949 3105 | hello.mateorobles@gmail.com
linkedin.com/in/automateon | github.com/SyberLabs | mateo.syberlabs.space

EDUCATION

Santa Clara University

Santa Clara, CA

B.S. Computer Science, Data Science Emphasis

June 2026

- Relevant coursework: Artificial Intelligence, Machine Learning, Algorithms, Programming Languages

EXPERIENCE

SyberLabs

Walnut Creek, CA

Founder and Software Engineer, open-source research and engineering studio

2026 – Present

- Built and open-sourced 4 products in 2026 (RISE, OmniOS, Relay, Dynamic Circulator) by setting product direction for all four, architecting three, and co-developing with a second engineer through code review and CI.
- Launched **RISE** at rise.syberlabs.io, an audiovisual reader that paces text against museum imagery and Web Audio, with a user-controlled photosensitivity toggle for rhythmic effects.
- Sought design critique on **RISE** from four HCI, typography and media-studies experts, and turned it into an ongoing redesign of reading pace, type scale and image-text pairing.
- Cut the **RISE** JavaScript bundle by **15.4 MB**, holding first load to about **59 KB** Brotli, by moving reading content into a SHA-256-verified data plane in a Vite app backed by Cloudflare Workers, PostgreSQL and Redis.
- Made every LLM answer traceable to its inputs by building **OmniOS** (in development), a local-first Next.js/TypeScript canvas where personas see only the live data wired into them (Polymarket, World Bank, OpenAlex, Hacker News), running keyless on local models via Ollama.
- Let OmniOS users add unfamiliar APIs without hand-written integrations by building a compiler that turns OpenAPI specs and **MCP** tool schemas into typed blocks, with approval-gated writes and idempotent execution that reports when a write's outcome is uncertain.
- Modeled adaptive crowd behavior in Dynamic Circulator with decaying preference memory and UCB edge learning, served through a **FastAPI** REST/WebSocket API to a React/Vite dashboard and covered by **100+ pytest** tests.

PROJECTS

Eidola | *Generative art series* | mateo.syberlabs.space

2025 – 2026

- Created **Eidola**, a series of seeded, reproducible generative-art engines in WebGL2 and Canvas 2D (RK4-integrated strange attractors, phase-interference fields, space-colonization growth), each with live palette and parameter controls.

SkyPredict V2 | *Machine learning course project, team of 3* | github.com/SyberLabs/skypredict-v2

2026

- Predicted U.S. flight delays before departure at **0.713 ROC-AUC**, 0.349 PR-AUC and **19.23-minute MAE** on a 1.16M-flight chronological holdout by building a leakage-aware LightGBM pipeline over **6.96M** BTS records.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS; familiar with C++, C, Go, C#

Creative Coding: WebGL2, Canvas 2D, Web Audio, generative and procedural systems, React, Vite

AI / LLM: LLM applications, AI agents, tool calling, Model Context Protocol (MCP), OpenAPI, Ollama, Anthropic and Gemini APIs, inference logging

Backend: FastAPI, Pydantic, REST, WebSockets, PostgreSQL, Redis, Cloudflare Workers, D1, Drizzle ORM

Testing / Infra: Git, GitHub Actions CI/CD, pytest, Vitest, Playwright, ESLint

ML / Data: PyTorch, scikit-learn, LightGBM, NumPy, SciPy, pandas, feature engineering, time-series, model evaluation